Fake News Generator and Detector

IBM PBEL (Project Based Experiential Learning)

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Introduction

The Problem

Digital misinformation spreads rapidly across platforms, undermining trust and influencing public opinion. Automatic detection is essential.

Our Solution

Combine GPT-2 (generative) to simulate fake news with BERT (discriminative) to detect it. Provides a controlled environment to test and harden classifiers.

Generative AI · NLP · Transformer Models — GPT-2 for generation, BERT for detection, Gradio UI

Project Objectives

Generate Synthetic News

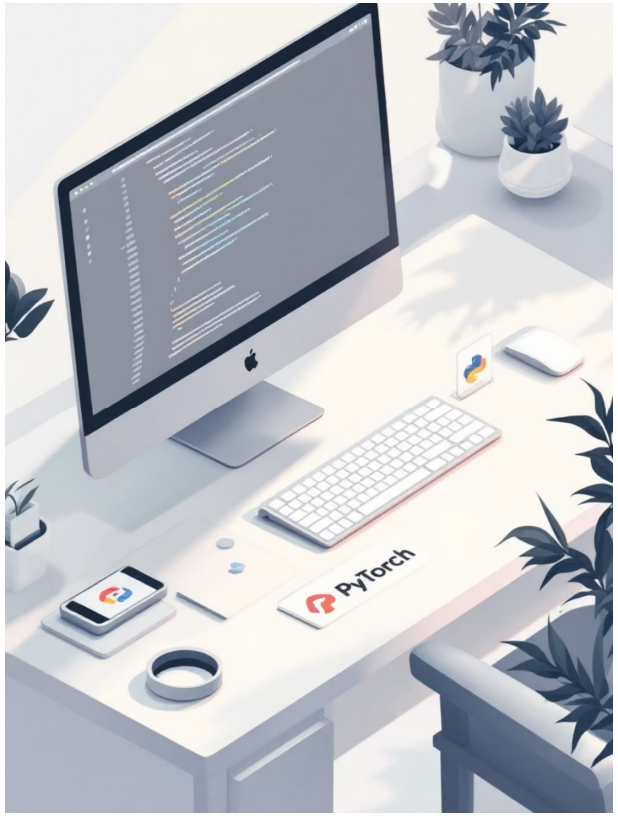
Use GPT-2 with top-k sampling and temperature=0.9 to create realistic fake news articles from prompts

Fine-tune BERT Detector

Train BERT-base-uncased over 5 epochs on labeled real/fake news data for binary classification

Interactive Gradio Dashboard

Build a split-screen UI with generation panel on left and detection panel on right for real-time testing



Technology Stack

- Python 3.8+: Core scripting and data pipelines
- PyTorch: Deep learning framework for model training
- Hugging Face Transformers: GPT-2 and BERT implementations
- Newspaper3k: Article scraping and text extraction from URLs
- Gradio: Interactive web UI with Glass theme
- Pandas & Scikit-learn: Data preparation and evaluation metrics

Device: CUDA GPU if available, else CPU

Generative Engine — GPT-2

GPT-2 uses causal language modeling to autoregressively generate fluent news-like paragraphs from short prompts. Configuration details:

Prompt Design

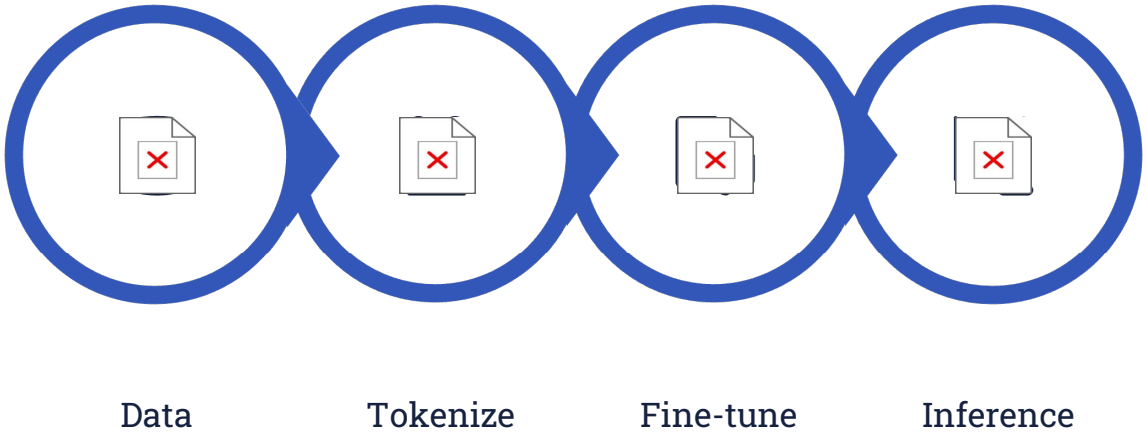
Contextual prompts guide generation and control coherence.

Sampling Controls

Temperature=0.9, top_k=50, top_p=0.95 balance novelty vs plausibility.

Ethical Constraints

Generated outputs are labeled and gated for research purposes only.



Discriminative Engine — BERT

Fine-tuned BERT-base-uncased for binary classification (Real vs. Fake). The pipeline includes: data preparation, tokenization, fine-tuning, and inference with confidence scores.

BERT analyzes word context bidirectionally, making it effective at detecting subtle linguistic cues in misinformation. Training: 5 epochs, batch size 2, cross-entropy loss with AdamW optimizer. Output: Softmax probabilities for Real/Fake classification.

Implementation Highlights

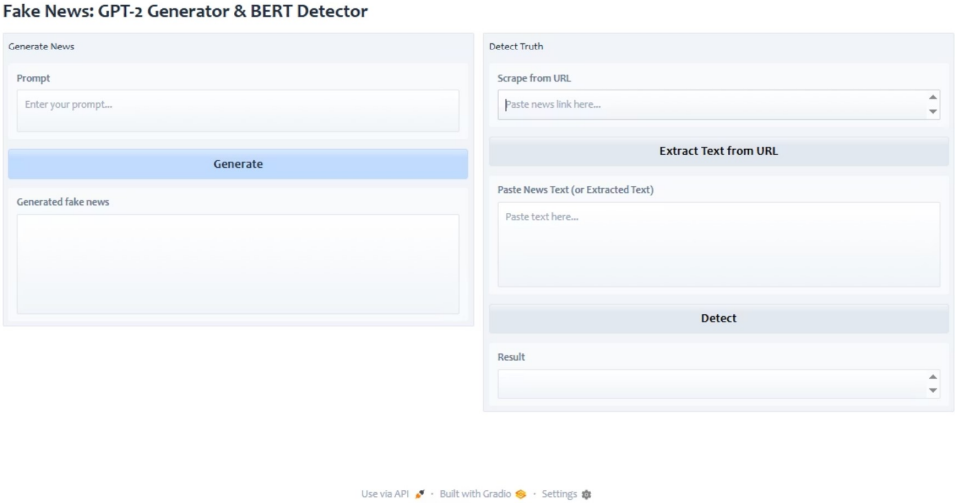
Dataset Preparation

Small labeled dataset created with Pandas: 4 sample news items labeled as Real (0) or Fake (1). Split into 80% training and 20% validation sets.

Gradio Dashboard

Split-screen interface: left panel for GPT-2 generation with prompt input, right panel for BERT detection with text input and confidence score output.

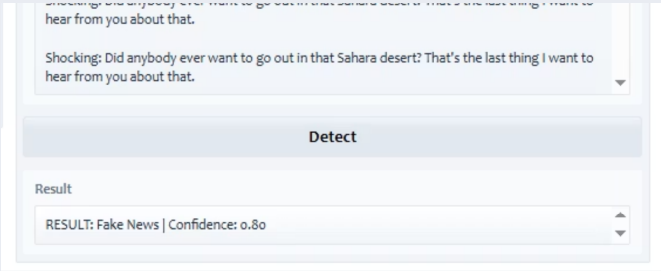
The Fake News Generator & Detector UI-



Experimental Results

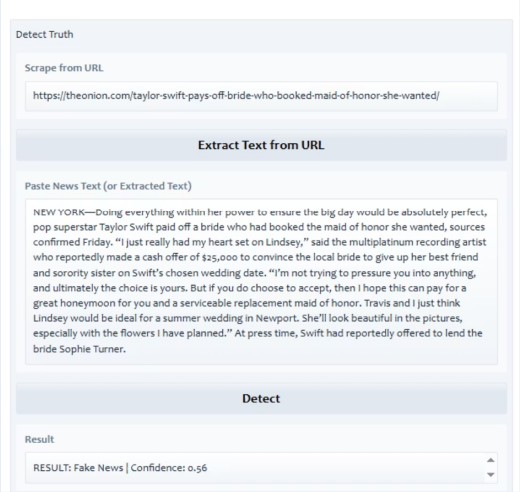
Summary of model behavior and reliability from controlled experiments:

Model Output



BERT produces calibrated probability scores via Softmax function. Predicted class (0=Real, 1=Fake) with confidence percentage displayed to user

Real-world Testing



Successfully extracts text from live news URLs using Newspaper3k and processes through detector for verification

Confidence scores guide threshold selection for actionable alerts.

Conclusion & Future Scope

Project Success

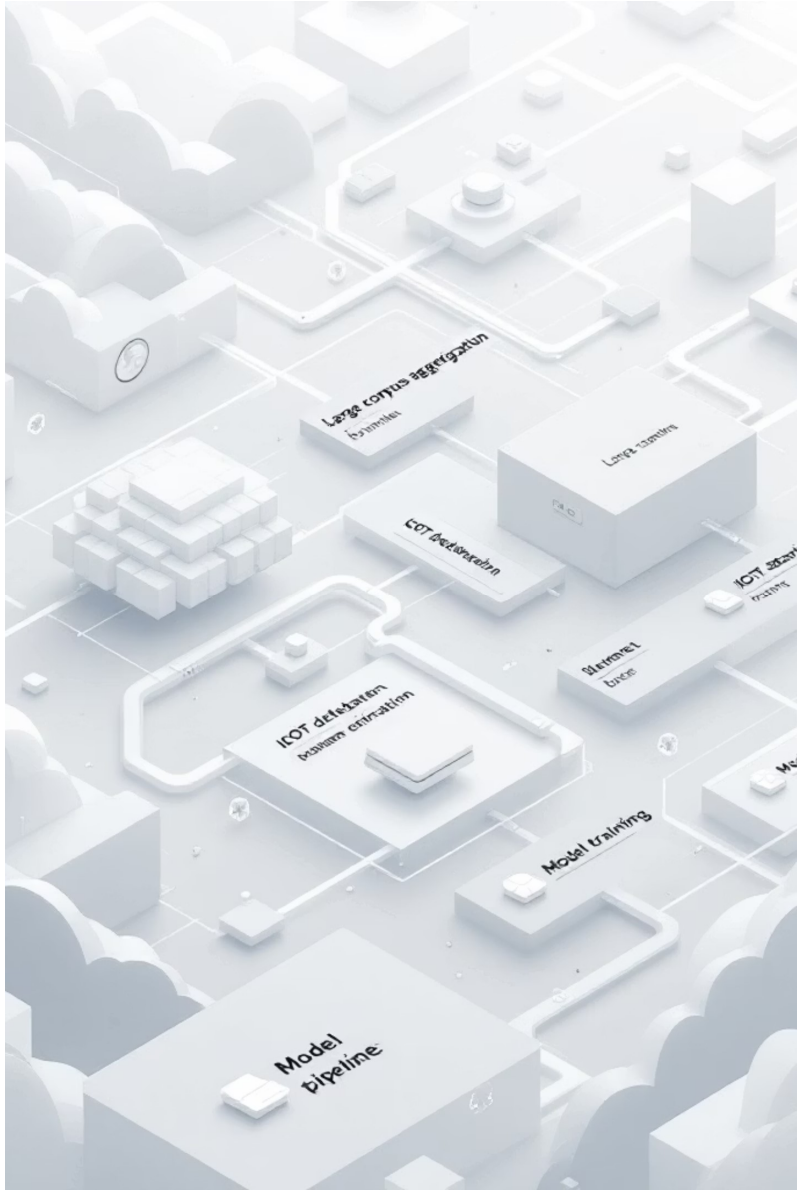
Successfully combined GPT-2 generation with BERT detection to create a controlled environment for testing fake news classifiers. Gradio UI enables real-time interaction.

Future Enhancements

Scale with larger datasets (ISOT), add explainable AI features (LIME), implement ensemble models, explore multimodal signals (images + text)

Deployment

Privacy-preserving logging, human-in-the-loop review, API rate limits, responsible demo access controls



References

Devlin, J., et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. https://arxiv.org/abs/1810.04805

Radford, A., et al. (2019). Language Models are Unsupervised Multitask Learners (GPT-2). https://openai.com/research/gpt-2

Hugging Face Transformers: https://huggingface.co/docs/transformers

Newspaper3k: https://newspaper.readthedocs.io

THANK YOU